

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: Project1
Study Case: SC

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Electrical Transient Analyzer Program

Short-Circuit Analysis

ANSI Standard

3-Phase, LG, LL, & LLG Fault Currents

1/2 Cycle Network

	Swing	V-Control	Load	Total
Number of Buses:	1	0	3	4

	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	1	0	0	1	0	1	3

	Synchronous Generator	Power Grid	Synchronous Motor	Induction Machines	Lumped Load	Total
Number of Machines:	0	1	2	0	0	3

System Frequency: 60.00
Unit System: English
Project Filename: Project1
Output Filename: C:\ETAP Course\ETAP projects\Project1\Project1\SC1.SA2S

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Adjustments

Tolerance	Apply Adjustments	Individual /Global	Percent
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		

Temperature Correction	Apply Adjustments	Individual /Global	Degree C
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
Bus1	SWNG	11.000	11.000	1	100.00	0.00
Bus2	Load	11.000	11.000	1	100.00	0.00
Bus3	Load	0.400	0.400	1	100.00	-30.00
Bus4	Load	0.400	0.400	1	100.00	-30.00

4 Buses Total

All voltages reported by ETAP are in % of bus Nominal kV.
Base kV values of buses are calculated and used internally by ETAP.

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Line/Cable/Busway Input Data

ohms or siemens per 1000 ft per Conductor (Cable) or per Phase (Line/Busway)

Line/Cable/Busway			Length		#/Phase	T (°C)	R1	X1	Y1	R0	X0	Y0
ID	Library	Size	Adj. (ft)	% Tol.								
Cable1	15NCUS3	95	328.1	0.0	1	75	0.0718055	0.0374904		0.1141708	0.0952256	

Line / Cable / Busway resistances are listed at the specified temperatures.

2-Winding Transformer Input Data

Transformer	Rating					Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	MVA	Prim. kV	Sec. kV	% Z	X/R	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
TRANS	2.000	11.000	0.400	6.00	7.10	0	0	0	0	0	6.00	Dyn	30.00

2-Winding Transformer Grounding Input Data

Transformer	Rating			Grounding									
	ID	MVA	Prim. kV	Sec. kV	Conn.	Primary			Secondary				
	ID	MVA	Prim. kV	Sec. kV	Type	Type	kV	Amp	ohm	Type	kV	Amp	ohm
TRANS		2.000	11.000	0.400	D/Y					Solid			

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Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
TRANS	2W XFMR	Bus2	Bus3	41.85	297.07	300.00	
BusDuct1	Bus Duct	Bus3	Bus4				
Cable1	Cable	Bus1	Bus2	1.95	1.02	2.20	

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Power Grid Input Data

Power Grid		Connected Bus		Rating		% Positive Seq. Impedance 100 MVA Base			Grounding	% Zero Seq. Impedance 100 MVA Base		
ID		ID		MVASC	kV	X/R	R	X	Type	X/R	R0	X0
U1		Bus1		250.000	11.000	10.00	3.98015	39.80149	Wye - Solid	10.00	3.980149	39.80149

Total Power Grids (= 1) 250.000 MVA

Synchronous Motor Input Data

Synchronous Motor			Rating (Base)			Positive Sequence Imp.					Grounding			Zero Seq. Imp.		
						Xd"					Conn.	Type	Amp	X/R	% R0	% X0
ID	Type	Qty	kVA	kV	RPM	X"/R	% R	Adj.	Tol.	% X'						
MOTOR 1	Motor	1	598.1	0.400	1800	21.15	0.727	15.385	0.0	23.077	Wye	Open		21.15	0.73	15.38
MOTOR 2	Motor	1	354.6	0.400	1800	15.62	0.985	15.385	0.0	23.077	Wye	Open		15.62	0.98	15.38

Total Connected Synchronous Motors (= 2): 952.7 kVA

SHORT- CIRCUIT REPORT

Fault at bus: Bus1

Prefault voltage = 11.000 kV
= 100.00 % of nominal bus kV (11.000 kV)
= 100.00 % of base kV (11.000 kV)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus1	Total	0.00	13.395	0.00	100.33	100.37	13.303	13.303	3.87E+000	3.90E+001	3.98E+000	3.98E+001
Bus2	Bus1	0.11	0.274	0.08	100.31	100.40	0.181	0.000	1.30E+002	1.91E+003		
U1	Bus1	100.00	13.122	100.00	100.00	100.00	13.122	13.303	3.98E+000	3.98E+001	3.98E+000	3.98E+001

Indicates fault current contribution is from three-winding transformers
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta- Y transformer

Fault at bus: Bus2

Prefault voltage = 11.000 kV
= 100.00 % of nominal bus kV (11.000 kV)
= 100.00 % of base kV (11.000 kV)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus2	Total	0.00	12.998	0.00	100.40	101.73	12.727	12.727	5.74E+000	4.00E+001	7.08E+000	4.24E+001
Bus1	Bus2	5.33	12.725	6.70	100.33	100.36	12.549	12.727	5.93E+000	4.08E+001	7.08E+000	4.24E+001
Bus3	Bus2	15.66	0.274	64.16	62.55	100.00	0.179	0.000	1.28E+002	1.91E+003		

Indicates fault current contribution is from three-winding transformers
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta- Y transformer

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Fault at bus: Bus3

Prefault voltage = 0.400 kV
= 100.00 % of nominal bus kV (0.400 kV)
= 100.00 % of base kV (0.400 kV)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus3	Total	0.00	51.194	0.00	100.63	101.49	50.125	50.125	3.52E+001	2.80E+002	4.19E+001	2.97E+002
Bus2	Bus3	87.91	42.297	94.14	100.00	94.15	44.317	50.125 *	4.78E+001	3.38E+002	4.19E+001	2.97E+002
MOTOR 1	Bus4	100.00	5.605	100.00	100.00	100.00	3.659	0.000	1.22E+002	2.57E+003		
MOTOR 2	Bus4	100.00	3.320	100.00	100.00	100.00	2.167	0.000	2.78E+002	4.34E+003		

Indicates fault current contribution is from three-winding transformers
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta- Y transformer

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Fault at bus: Bus4

Prefault voltage = 0.400 kV
= 100.00 % of nominal bus kV (0.400 kV)
= 100.00 % of base kV (0.400 kV)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus4	Total	0.00	51.194	0.00	100.63	101.49	50.125	50.125	3.52E+001	2.80E+002	4.19E+001	2.97E+002
MOTOR 1	Bus4	100.00	5.605	100.00	100.00	100.00	3.659	0.000	1.22E+002	2.57E+003		
MOTOR 2	Bus4	100.00	3.320	100.00	100.00	100.00	2.167	0.000	2.78E+002	4.34E+003		
Bus2	Bus3	87.91	42.297	94.14	100.00	94.15	44.317	50.125 *	4.78E+001	3.38E+002	4.19E+001	2.97E+002

Indicates fault current contribution is from three-winding transformers
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta- Y transformer

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Short-Circuit Summary Report

1/2 Cycle - 3-Phase, LG, LL, & LLG Fault Currents

Prefault Voltage = 100 % of the Bus Nominal Voltage

Bus		3-Phase Fault			Line-to-Ground Fault			Line-to-Line Fault			*Line-to-Line-to-Ground		
ID	kV	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.
Bus1	11.000	1.324	-13.330	13.395	1.318	-13.237	13.303	11.544	1.147	11.601	-12.200	5.426	13.352
Bus2	11.000	1.847	-12.866	12.998	1.908	-12.583	12.727	11.143	1.600	11.257	-12.124	4.556	12.952
Bus3	0.400	6.399	-50.793	51.194	6.518	-49.700	50.125	43.988	5.542	44.335	-47.299	18.783	50.892
Bus4	0.400	6.399	-50.793	51.194	6.518	-49.700	50.125	43.988	5.542	44.335	-47.299	18.783	50.892

All fault currents are symmetrical (1/2 Cycle network) values in rms kA.

* LLG fault current is the larger of the two faulted line currents.

Sequence Impedance Summary Report

Bus		Positive Seq. Imp. (ohm)			Negative Seq. Imp. (ohm)			Zero Seq. Imp. (ohm)			Fault Zf (ohm)		
ID	kV	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance
Bus1	11.000	0.04687	0.47179	0.47411	0.04687	0.47179	0.47411	0.04816	0.48160	0.48400	0.00000	0.00000	0.00000
Bus2	11.000	0.06943	0.48363	0.48859	0.06943	0.48363	0.48859	0.08562	0.51284	0.51994	0.00000	0.00000	0.00000
Bus3	0.400	0.00056	0.00448	0.00451	0.00056	0.00448	0.00451	0.00067	0.00475	0.00480	0.00000	0.00000	0.00000
Bus4	0.400	0.00056	0.00448	0.00451	0.00056	0.00448	0.00451	0.00067	0.00475	0.00480	0.00000	0.00000	0.00000